

1 Question and response

Which surface and form properties explain where night air is warmer or cooler than the district network?

Response: $\Delta T = T_{\text{site}} - \text{network median}$, 18:00–06:00 HST mean per sensor-night

2 Measurement

iButton loggers in ventilated cup shields, $\approx 1.5\text{m}$, 30-min interval
Honolulu 36 sites · 'Ewa 38 sites
Nov 2024 – Jan 2025
Random-point site selection ($\geq 50\text{m}$ apart)

3 Processing

Readings snapped to 30-min bins
Network median per bin and district
ECMWF wind and cloud joined
Night = 18:00–06:00 (diurnal date starts 06:00)

6 Scale choice

Full model at each radius (ML AIC) pooled, Honolulu, 'Ewa
→ 100m adopted (best or within 2 AIC in all three)
Collinearity screen (r , VIF)

5 Spatial predictors

C-CAP 1m: impervious, tree, water
FEMA footprints: building fraction, height, canyon H/W, sky view factor
Distance to the shoreline (GSHHG coastline database)
Radii 50 / 100 / 200m

4 Night selection

Calm and clear: wind $< 10\text{km h}^{-1}$ and cloud $< 25\%$ in $\geq 75\%$ of night bins; ≥ 20 sensors; ≥ 18 of 24 bins per sensor-night
→ 7 Honolulu + 4 'Ewa nights, 400 sensor-nights

7 Mixed-effects models (lme4)

Random intercept per site
Null → single predictors → full (8 predictors + district)
District models, best subsets
District $\times$ predictor interactions

8 Checks

Threshold sensitivity (re-select nights, refit); calm-core refit
Terrain (elevation) check
Site checks: HNL08, missing heights, sky view factor
Leave-one-site-out CV

9 Interpretation

Effects in natural units (per 10 pp impervious, per m of height)
BLUP maps of unexplained heat
Absolute comparison on the two shared nights